In [1]:
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import cv2  # Pragya Singhal, Cu24220050 , 43

import matplotlib.pyplot as plt
import numpy as np
image_path = 'img.jpeg'
img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
if img is None:
  print(
      '[INFO] Image not found. Creating a synthetic test pattern for'
      ' demonstration.'
  )
  img = np.zeros((300, 300), dtype=np.uint8)
  cv2.putText(
      img,
      'TEST',
      (50, 160),
      cv2.FONT_HERSHEY_SIMPLEX,
      3,
      255,
      6,
      cv2.LINE_AA,
  )
  cv2.circle(img, (150, 150), 60, 200, -1)

# # Pragya Singhal, Cu24220050 , 43
dft = cv2.dft(np.float32(img), flags=cv2.DFT_COMPLEX_OUTPUT)
dft_shift = np.fft.fftshift(dft)

# Calculate magnitude spectrum for visualization    # Pragya Singhal, Cu24220050
magnitude_spectrum = 20 * np.log(
    cv2.magnitude(dft_shift[:, :, 0], dft_shift[:, :, 1]) + 1
)
rows, cols = img.shape
crow, ccol = rows // 2, cols // 2
# Low-Pass Filter (LPF)         # Pragya Singhal, Cu24220050 , 43
r_lpf = 30
mask_lpf = np.zeros((rows, cols, 2), np.uint8)
cv2.circle(mask_lpf, (ccol, crow), r_lpf, (1, 1), -1)
fshift_lpf = dft_shift * mask_lpf
f_ishift_lpf = np.fft.ifftshift(fshift_lpf)
img_back_lpf = cv2.idft(f_ishift_lpf)
img_back_lpf = cv2.magnitude(img_back_lpf[:, :, 0], img_back_lpf[:, :, 1])

# High-Pass Filter (HPF)    # Pragya Singhal, Cu24220050 , 43
r_hpf = 30
mask_hpf = np.ones((rows, cols, 2), np.uint8)
cv2.circle(mask_hpf, (ccol, crow), r_hpf, (0, 0), -1)
fshift_hpf = dft_shift * mask_hpf
f_ishift_hpf = np.fft.ifftshift(fshift_hpf)
img_back_hpf = cv2.idft(f_ishift_hpf)
img_back_hpf = cv2.magnitude(img_back_hpf[:, :, 0], img_back_hpf[:, :, 1])

# Visualization   # Pragya Singhal, Cu24220050 , 43
plt.figure(figsize=(12, 8))
plt.subplot(2, 3, 1), plt.imshow(img, cmap='gray'), plt.title(
    'Original Image'
), plt.axis('off')
plt.subplot(2, 3, 2), plt.imshow(magnitude_spectrum, cmap='gray'), plt.title(
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    'Magnitude Spectrum'
), plt.axis('off')
plt.subplot(2, 3, 3), plt.imshow(img_back_lpf, cmap='gray'), plt.title(
    'Low-Pass Filtered'
), plt.axis('off')
plt.subplot(2, 3, 4), plt.imshow(img_back_hpf, cmap='gray'), plt.title(
    'High-Pass Filtered'
), plt.axis('off')
plt.subplot(2, 3, 5), plt.imshow(mask_lpf[:, :, 0] * 255, cmap='gray'), plt.titl
    'LPF Mask'
), plt.axis('off')
plt.subplot(2, 3, 6), plt.imshow(mask_hpf[:, :, 0] * 255, cmap='gray'), plt.titl
    'HPF Mask'
), plt.axis('off')
plt.tight_layout()
plt.show()
```

